

Research paper

Metagenomic study focusing on antibiotic resistance genes from the sediments of River Yamuna



Basanta Kumar Das^{a,*,1}, Bijay Kumar Behera^{a,*,1}, Hirak Jyoti Chakraborty^a, Prasenjit Paria^a, Aditi Gangopadhyay^b, Ajaya Kumar Rout^a, Kausalya Kumari Nayak^c, Pranaya Kumar Parida^a, Anil Rai^d

^a ICAR-Central Inland Fisheries Research Institute, Barrackpore, Kolkata 700120, India

^b Department of Chemical Technology, University of Calcutta, Kolkata 700009, West Bengal, India

^c KBDV College, Nirakarpur, Odisha 752019, India

^d Centre for Agricultural Bioinformatics, ICAR-Indian Agricultural Statistics Research Institute, Library Avenue, PUSA, New Delhi 110012, India

ARTICLE INFO

Keywords:

Antibiotic resistance genes
Sediment metagenome
Polluted riverine ecosystem
Yamuna River
Resistome

ABSTRACT

Antibiotic resistance is one of the major health concerns of the present century. The direct discharge of urban sewage, hospital effluents, and pharmaceutical wastes increases the concentration of antibiotics in riverine ecosystems. This provides selection pressure for the development of novel antibiotic-resistant strains. In this study, metagenomics approach was employed for constructing a comprehensive profile of the Antibiotic Resistance Genes (ARGs) identified in the sediments of the Yamuna River. A total of 139 ARGs were identified from 39 microbial species. Abundance analysis revealed that, aminoglycoside, beta-lactam, macrolide, and tetracycline resistance genes were highly abundant in the sediment samples obtained from the Yamuna River. The evolutionary relationships among the ARGs were studied by phylogenetic analyses, which revealed that, the identified resistome comprised eight clusters. Network analysis was performed for investigating the broad-spectrum profiles of the ARGs and their enrichment in different biological functions and pathways. Protein-protein interaction (PPI) analyses revealed that, 76, 36, 18, and 5 Gene Ontology (GO)-terms were significantly enriched in Biological process, Molecular Function, Cellular Component, and KEGG Pathways analysis, respectively. The present study elucidates the ecology of microbial antibiotic resistance in the riverine ecosystem of the Yamuna River and provides novel insights into the environmental hotspots that are amenable to the emergence of ARGs in the contaminated riverine hydrosphere.

1. Introduction

The excessive use of antibiotics in the agricultural and the health sectors over the last few decades has resulted in the emergence of a new bacterial population carrying multiple antibiotic resistant genes (ARGs) (Xu et al., 2015; Guo et al., 2017). At present, antibiotics and ARGs are being increasingly regarded as environmental pollutants. ARGs are naturally present in numerous bacterial species and the presence of such genes has been confirmed in remote and ancient soils and caves (Garner et al., 2016). The increasing concentration of antibiotics in the

environment provides a selection pressure which promotes the growth of resistant bacterial strains (Tello et al., 2012). The presence of antibiotics in the environment facilitates the acquisition of ARGs in different bacterial species by horizontal gene transfer viz. mobile genetic elements, including transposons, plasmids, and lysogenic bacteriophages (Mullany, 2014). The selective pressure together with horizontal gene transfer acts as a driving force for the emergence of antibiotic-resistant pathogenic bacteria (Zeng and Lin, 2017). The emergence of antibiotic-resistant pathogenic microbes is a serious threat to human health. The evolution of multidrug resistant (MDR)

Abbreviations: ARGs, Antibiotic Resistance Genes; MDR, Multidrug resistant; gDNA, Genomic DNA; ng, Nanogram; bp, Base pair; ORF, Open reading frame; AGR, Aminoglycoside resistance; BLR, beta-lactam resistance; FR, Fluoroquinolone resistance; MR, Macrolide resistance; LR, Lincosamide resistance; PR, Phenicol resistance; RR, Rifampicin resistance; SR, Sulphonamide resistance; TER, Tetracycline resistance; TRR, Trimethoprim resistance; NDM, New Delhi metallo-beta-lactamase; VIM, Verona Integron-encoded Metallo-beta-lactamase; KPC, Klebsiella pneumoniae carbapenemase

* Corresponding authors.

E-mail addresses: basanta.das@icar.gov.in (B.K. Das), beherabk18@yahoo.co.in (B.K. Behera).

¹ Authors equally contributed to this work.

<https://doi.org/10.1016/j.gene.2020.144951>

Received 15 March 2020; Received in revised form 26 June 2020; Accepted 13 July 2020

Available online 16 July 2020

0378-1119/ © 2020 Elsevier B.V. All rights reserved.

pathogenic superbugs in the environment traces back to the pre-antibiotic era (D'Costa et al., 2011; Martínez, 2008). A-resistant pathogenic bacteria are responsible for 700,000 deaths annually, according to a report by the World Health Organization published in 2018 (O'Neill, 2018). A recent study reported that, 10 million deaths will be caused by antibiotic-resistant pathogenic bacteria in the year 2050, and 100 trillion USD from the global economic output will be lost if substantial efforts are not made to control this threat (O'Neill, 2016; WHO, 2017).

Numerous studies have reported the presence of different antibiotics in the soil, sediments, and water environments, globally (Guo et al., 2017). The direct discharge of hospital effluents, pharmaceutical waste, and sewage into rivers results in the accumulation of antibiotics in the riverine ecosystem. These waterbodies serve as a reservoir of antibiotics and promote the growth of antibiotic-resistant bacteria (Bengtsson-Palme et al., 2014). Bacteria use different mechanisms, including efflux pumps, enzymatic inactivation or modification, and the induction of mutations in the target gene, for surviving against antibiotics (Mullany, 2014). It is difficult to identify ARGs from the environment and elucidate the mechanisms underlying antimicrobial resistance by classical experimental methods within a short period. Culture-independent metagenomics is a comprehensive strategy for identifying and studying the antibiotic resistome from environmental samples. The combination of computational metagenomics with high-throughput sequencing platforms has significantly advanced the field of antimicrobial resistance.

The Yamuna River is one of the mythological rivers in India with historical importance. Several important cities, including Delhi, Mathura, and Agra, are situated on the banks of the Yamuna River. The flow of the Yamuna is being gradually restricted and increasingly polluted by the enormous discharge of city sewage and industrial wastewater from New Delhi and its peripheral regions (Sahu and Sohoni, 2014). The low flow rate combined with the high quantities of organic compounds and antibiotics served as a natural bioreactor for the growth of antibiotic-resistant microorganisms. This study aimed to characterise the ARGs in the Yamuna River by constructing a comprehensive profile of the ARGs present in the sediments of the Yamuna River. The ARGs identified herein conferred resistance to 11 antibiotics, namely, aminoglycosides, beta-lactams, fluoroquinolones, macrolides, macrolide lincosamides with streptogramin B (MLS), lincosamides, phenicols, rifampicins sulphonamides, tetracyclines, and trimethoprim antibiotics. A statistical metagenomic study was conducted for analysing the location-specific abundance of ARGs. The evolutionary relationships among the different ARGs were determined by meta-phylogenetic analyses. Additionally, functional metagenomic analysis was performed for predicting the functionome of the identified antibiotic-resistant microbes in the riverine ecosystem of Yamuna.

2. Material and methods

2.1. Sampling, gDNA extraction, and library preparation

The Wazirabad barrage and Okhla barrage are two important barrages in the Yamuna River that restrict the flow of the river in the Delhi segment. Discharges from seventeen sewage drains of Delhi and Najafgarh drain into the Yamuna River, downstream of the Wazirabad barrage. Domestic and industrial wastewater from East Delhi, Noida, and Sahibabad flow into the Yamuna River downstream of the Okhla barrage, making this the most polluted segment of the river (Sahu and Sohoni, 2014; Misra 2010).

Three different sites along the Yamuna River, namely, Wazirabad (28°42.39' N 77°13.57' E), Okhla barrage (28°32.51' N 77°18.30' E), and Faizupur Khaddar (28°18.43' N 77°27.52' E) near New Delhi were selected for sediment collection (Fig. 1). The samples collected from the sampling sites at Wazirabad, Okhla barrage, and Faizupur Khaddar are hereafter referred to as Y1S1, Y2S2, and Y3S3, respectively. The sampling strategy comprised the collection of random sediment samples from five different spots around one sampling site. All the sediment

samples were homogenized, mixed, and pooled together as one sample. The sediment samples were kept in a sealed sterile plastic container and transported on the ice at 4 °C, and subsequently stored at −80 °C until further processing. All the tools and containers that had been used for sampling were sterilised before sample collection. The environmental DNA was extracted from the sediment samples using a gDNA isolation kit (Nucleospin Soil). Sufficient quantities of high-quality DNA were isolated for genomic library preparation. Following genomic DNA extraction, the quality of the extracted DNA was checked using a Nanodrop 2000 and Qubit® 3.0 Fluorometer. An Illumine TruSeq Nano DNA Library Prep Kit (illumina) was used for the construction of a paired-end sequencing library. Approximately 200 ng of genomic DNA was fragmented by Covaris M220 to produce a mean fragment allocation of 350 bp. Covaris shearing produces dsDNA fragments with 3' or 5' overhangs. The fragments thus obtained were subjected to end-repair, and the products were amplified by PCR using the index primer as described in the kit. The PCR-enriched libraries were investigated using a 4200 Tape Station system (Agilent Technologies) using a D1000 Screen tape.

2.2. Sequencing, quality control, and de-novo assembly

After obtaining the Qubit concentration for the libraries and the mean peak size from the Agilent Tape Station profile, the PE Illumina libraries were loaded into NextSeq 500 for cluster generation and sequencing. The sequenced raw data was processed to obtain high quality Clean read Using Trimmomatic V0.35 to remove adapter sequences, ambiguous read (reads with unknown nucleotides "N" larger than 5%) and low quality sequences (reads with more than 10% quality threshold {QV} < 20 Phred Score). A minimum length of 100 nt was applied after trimming. The filtered high quality reads for each sample were then assembled into scaffolds using the CLC Genomics Workbench, version 8.5.1. The details read quality statistics and assembly statistics were given in [Supplementary Tables S2 and S3](#).

2.3. Sequence information

The metagenomic sequences reported in this study have been submitted to the NCBI-SRA database under SRA Accession Numbers are PRJNA531081, PRJNA531090 and PRJNA531097 for the sediment samples Y1S1, Y2S2, and Y3S3, respectively, collected from the Yamuna River.

2.4. Detection of ARGs from the sediment metagenome of Yamuna

The ARGs in the sediment metagenome of the Yamuna River was determined using the server-based web interface of ResFinder, version 3.0 (Zankari et al., 2012). Using the ResFinder tools, we identified ARGs conferring resistance to aminoglycoside, beta-lactam, colistin, fluoroquinolone, fosfomycin, fusidic-acid, glycopeptide, MLS, nitroimidazole, oxazolidinone, phenicol, rifampicin, sulphonamide, tetracycline, and trimethoprim antibiotics in the sediment metagenome of the Yamuna River. These ARGs were subsequently subjected to a BLAST search against the ARG database of ResFinder, version 3.0 for identifying the homologous genes (Altschul et al., 1990). Identity of 80% was considered to be the threshold for reporting a match between a gene in the ResFinder database and the input genome.

2.5. Statistical data analysis for the identified ARGs

The number of ARGs in the different microbial species was identified and a relational matrix of the number of ARGs and the corresponding species was subsequently prepared for dataset analysis. A heatmap of the observed number of resistant microbes and the types of antibiotics against which the microbes developed resistance was prepared using Clustvis (Metsalu and Vilo, 2015). Abundance analysis was

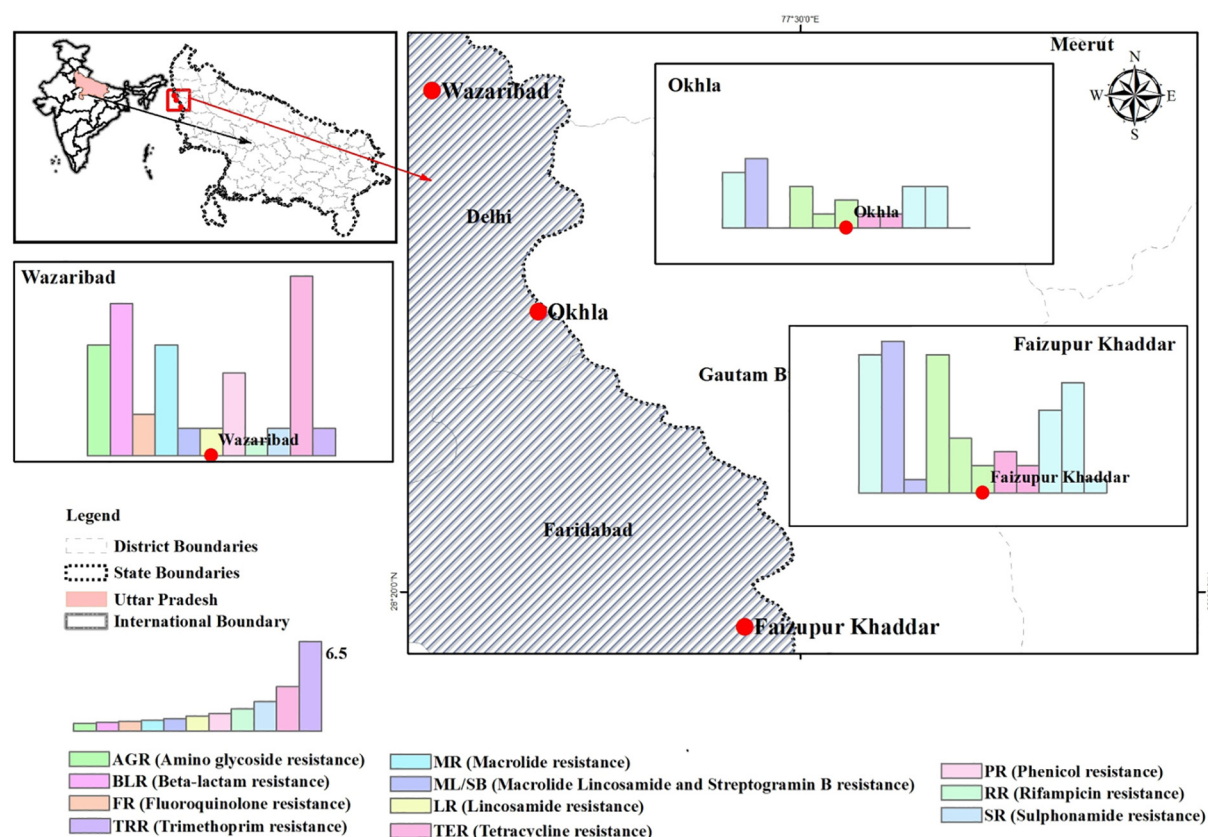


Fig. 1. Map showing the sediment sampling sites. Sediments were collected from three sampling site of river Yamuna, namely Wazaribad, Okhla and Faizupur Khaddar further different ARGs types were depicted. This map was prepared in ARC-GIS.

performed by calculating the in-between abundance of ARGs among the number of ARGs-like reads, and dividing by the total number of sequencing reads in the same sample (Chen et al., 2018).

2.6. Phylogenetic analyses of ARGs in the Yamuna River sediment

The evolutionary relationships among the ARGs identified in the sediment samples collected from the Yamuna River was investigated by phylogenetic analysis using the CLC Genomics Workbench, version 8.5.1. The phylogenetic tree was constructed using neighbor-joining approach, and the bootstrap value was set to 1000.

2.7. Prediction of resistance gene-encoded open reading frames (ORFs) and network analysis

The functional portion of the gene in the identified resistome was determined by predicting the resistance gene-encoded ORFs using the CLC Genomics Workbench, version 8.5.1. The resistance gene-encoded ORFs were subsequently subjected to functional metagenomics analyses using the STRING web interface. STRING (<https://string-db.org/>) is a protein network analysis database of pre-computed data about gene and protein interactions. The direct interactions are mediated by direct binding, and the indirect interactions are predicted from functional similarity. The functional partners involved in the interactions were ranked according to the estimated confidence scores obtained from the prediction algorithms based on genomic information. The probabilistic confidence scores were scored based on a special scoring framework in STRING, which is based on the relationships in a common reference set. The interactions are integrated with sources such as genomic context, high throughput experimental data, literature survey, co-expressed gene analysis, and database data mining (Franceschini et al., 2012;

Szklarczyk et al., 2010; Szklarczyk et al., 2016).

3. Results

3.1. Identification of the resistome in the sediment metagenome of Yamuna

BLAST search against the ARG database of ResFinder version 3.0 revealed that, the three sediment metagenomes from samples Y1S1, Y2S2, and Y3S3, collected from three different locations of the Yamuna River, contained a total of 58, 23, and 58 ARGs, respectively. The microbes were found to be resistant to 11 different types of antibiotics, namely, aminoglycoside, beta-lactam, fluoroquinolone, macrolide, MLS, lincosamide, phenicol, rifampicin, sulphonamide, tetracycline, and trimethoprim antibiotics. The resistant genes, the types of antibiotics to which they conferred resistance, the contigs ID, the total number of observed values, and the gene identity scores of the producing organism are depicted in Table 1.

3.2. Occurrence and abundance analysis of ARGs

A heatmap of the number of observed resistant microbial species was prepared concerning the different types of antibiotics against which resistance developed (Fig. 2). Heatmap analysis revealed that out of the 39 different resistance microbes, 10 microbial species, including *Pseudomonas aeruginosa*, *Shigella sonnei*, *Escherichia coli*, *Enterococcus faecium*, *Vibrio cholerae*, *Streptococcus pneumoniae*, *Acinetobacter baumannii*, *Streptococcus suis*, *Salmonella enterica*, and some uncultured bacteria, showed resistance against multiple drugs.

Relative abundance analysis of the ARGs identified from the three different sites along the Yamuna River revealed the 11 types of ARGs identified herein, namely, AGR (aminoglycoside resistance), BLR (beta-

Table 1

Types of antibiotic resistance genes identified from the sediment metagenome of river Yamuna.

Types	Resistance gene	Contigs	Total no	Identity	Name/Accession no.
Y1s1	Amino glycoside resistance	aadA1	8	99.75	<i>Pseudomonas aeruginosa</i> (JQ414041)
		aac(6)-IIa		99.23	<i>Pseudomonas aeruginosa</i> (M29695)
		ant(2'')-Ia		100	<i>Shigella sonnei</i> (M86913)
		aadA5		100	<i>Escherichia coli</i> (AF137361)
		aph(6)-Id		100	<i>Escherichia coli</i> (M28829)
		ant(6)-Ia		99.77	<i>Enterococcus faecium</i> (KF421157)
		aadA2		100	<i>Escherichia coli</i> (X68227)
		aacA4		99.78	<i>Escherichia fergusonii</i> (KM278199)
		blaAER-1		92.33	<i>Aeromonas hydrophila</i> (U14748)
		blaOXA-45		84.33	<i>Pseudomonas aeruginosa</i> (AJ519683)
	Beta-lactam resistance	blaNPS	11	99.87	<i>Pseudomonas aeruginosa</i> (AY027589)
		cfxA3		99.75	<i>Capnocytophaga ochracea</i> (AF472622)
		blaOXA-10		100	<i>Pseudomonas aeruginosa</i> PA96 (EU886981)
		blaOXA-21		98.55	<i>Aeromonas hydrophila</i> (DQ993182)
		blaOXA-347		100	uncultured bacterium [H G] (JN086160)
		blaOXA-1		99.04	Plasmid RGN238 [<i>E.coli</i>] (J02967)
		blaGES-5		100	<i>Pseudomonas aeruginosa</i> (DQ236171)
		blaVEB-1		100	<i>Aeromonas media</i> (HM370393)
		blaVEB-1		93.1	<i>Aeromonas media</i> (HM370393)
		qnrVC4		100	<i>Aeromonas caviae</i> (GQ891757)
	Fluoroquinolone resistance	aac(6')-Ib-cr	3	99.58	<i>Escherichia coli</i> (DQ303918)
		qnrVC1		100	<i>Vibrio cholerae</i> (EU436855)
		ere(A)		98.85	<i>Escherichia coli</i> (AY183453)
	Macrolide resistance	mef(A)	8	92.08	<i>Streptococcus pneumoniae</i> (AF376746)
		erm(F)		99.75	<i>Bacteroides fragilis</i> (M17808)
		mph(E)		100	<i>Acinetobacter baumannii</i> (EU294228)
		mph(A)		100	<i>Escherichia coli</i> (D16251)
		erm(A)		84.09	<i>Streptococcus suis</i> D9 (EU348758)
		mef(A)		90.38	<i>Streptococcus pneumoniae</i> (AF376746)
		mef(B)		87.92	<i>Escherichia coli</i> (FJ196385)
		msr(D)		91.46	<i>Streptococcus pneumoniae</i> (AF274302)
		msr(E)		100	<i>Acinetobacter baumannii</i> (EU294228)
		lnu(F)		98.67	<i>Escherichia coli</i> (AJ561197)
	Macrolide, Lincosamide and Streptogramin B resistance	lnu(B)	2	98.96	<i>Enterococcus faecium</i> (AJ238249)
		cmlA1		99.13	<i>Escherichia coli</i> (AB212941)
	Lincosamide resistance	catB4	6	100	<i>Escherichia coli</i> (EU935739)
		catQ		99.7	<i>Clostridium perfringens</i> (M55620)
		cat		95.06	<i>Campylobacter coli</i> (M35190)
		catB4		86.14	<i>Escherichia coli</i> (EU935739)
		floR		85.03	<i>Salmonella enterica</i> (AF118107)
		ARR-2		100	<i>Acinetobacter baumannii</i> (HQ141279)
		sul2		100	<i>Acinetobacter bereziniae</i> (GQ421466)
		sul1		99.88	<i>Klebsiella pneumoniae</i> (DQ143913)
		tet(33)		98.78	<i>Trueperella pyogenes</i> (AY255627)
		tet(G)		94.13	<i>Pseudomonas sp.</i> (AF133139)
	Sulphonamide resistance	tet(G)	13	100	<i>Pseudomonas sp.</i> (AF133139)
		tet(C)		99.92	uncultured bacterium (NC.003213)
		tet(M)		99.68	<i>Enterococcus faecalis</i> (X04388)
		tet(W)		99.57	<i>Streptococcus suis</i> (FN396364)
		tet(32)		100	<i>Escherichia coli</i> (EU722333)
		tet(36)		90.89	<i>Bacteroides coprosuis</i> (AJ514254)
		tet(O)		99.74	<i>Campylobacter jejuni</i> (M18896)
		tet(A)		100	<i>Acinetobacter baumannii</i> (AY196695)
		tet(40)		99.35	uncultured bacterium (FJ158002)
		tet(Q)		99.74	<i>Prevotella ruminicola</i> (L33696)
	Trimethoprim resistance	tet(39)	2	100	<i>Acinetobacter baumannii</i> (KT346360)
		dfrA32		100	<i>Salmonella enterica</i> (GU067642)
		dfrA1		100	<i>Escherichia coli</i> (X00926)

(continued on next page)

Table 1 (continued)

Types	Resistance gene	Contigs	Total no	Identity	Name/Accession no.
Y2s2	Aminoglycoside resistance	aadA24	4	93.57	<i>Salmonella enterica subsp.</i> (AM711129)
		aph(6)-Id		100	<i>Escherichia coli</i> (M28829)
		aac(6')-31		86.39	<i>Acinetobacter baumannii</i> (AJ640197)
		aph(3'')-Ib		100	<i>Shigella flexneri</i> (AF321551)
	Beta-lactam resistance	blaVEB-1	5	100	<i>Aeromonas media</i> (HM370393)
		blaNPS		99.87	<i>Pseudomonas aeruginosa</i> (AY027589)
		blaOXA-10		100	<i>Pseudomonas aeruginosa</i> PA96 (EU886981)
		blaGES-5		100	<i>Pseudomonas aeruginosa</i> (DQ236171)
	Macrolide, Lincosamide and Streptogramin B resistance	blaOXA-21	1	98.19	<i>Aeromonas hydrophila</i> (DQ993182)
		msr(D)		91.45	<i>Streptococcus pneumoniae</i> (AF227520)
	Macrolide resistance	ere(A)	3	98.85	<i>Escherichia coli</i> (AY183453)
		erm(F)		99.66	<i>Bacteroides fragilis</i> (M17808)
		mph(E)		100	<i>Acinetobacter baumannii</i> (EU294228)
	Lincosamide resistance	lnu(F)	2	100	<i>Escherichia coli</i> (AJ561197)
		lnu(F)		83.56	<i>Escherichia coli</i> (AJ561197)
	Phenicol resistance	catQ	1	99.24	<i>Clostridium perfringens</i> (M55620)
	Rifampicin resistance	ARR-2	1	100	<i>Acinetobacter baumannii</i> (HQ141279)
	Sulphonamide resistance	sul3	3	93.18	<i>Escherichia coli</i> (AJ459418)
		sul2		100	<i>Acinetobacter bereziniae</i> (GQ421466)
		sul1		100	<i>Escherichia coli</i> (AY224185)
	Tetracycline resistance	tet(C)	3	99.92	uncultured bacterium (NC_003213)
		tet(A)		100	<i>Aeromonas salmonicida</i> (AJ517790)
		tet(M)		99.57	<i>Streptococcus pneumoniae</i> (FR671418)

(continued on next page)

Table 1 (continued)

Types	Resistance gene	Contigs	Total no	Identity	Name/Accession no.
Y3s3	Aminoglycoside resistance	aadA2	10	99.62	<i>Escherichia coli</i> (JQ364967)
		aadA11		99.88	<i>Pseudomonas aeruginosa</i> (AY144590)
		ant(2'')-Ia		100	uncultured bacterium (X04555)
		aac(6)-IIa		99.64	<i>Pseudomonas aeruginosa</i> (M29695)
		aadA24		90.72	<i>Salmonella enterica</i> (AM711129)
		aadA5		100	<i>Escherichia coli</i> (AF137361)
		aph(6)-Id		100	<i>Escherichia coli</i> (M28829)
		ant(6)-Ia		100	<i>Enterococcus faecium</i> (KF421157)
		aadA1		99.62	<i>Pseudomonas aeruginosa</i> (JQ414041)
		aadA10		99.7	<i>Pseudomonas aeruginosa</i> (U37105)
	Beta-lactam resistance	blaOXA-372	11	92.94	<i>Citrobacter freundii</i> (KJ746496)
		blaOXA-4		100	<i>Citrobacter freundii</i> (AY162283)
		blaOXA-9 blaVEB-1		99.81	<i>Enterobacter sp.</i> (JF703130)
		blaNPS		100	<i>Aeromonas media</i> (HM370393)
		blaOXA-10		99.87	<i>Pseudomonas aeruginosa</i> (AY027589)
		blaOXA-347		100	(EU886981)
		blaTEM-1B		100	Uncultured bacterium (JN086160)
		blaGES-5		100	<i>Leclercia adecarboxylata</i> (JF910132)
		blaOXA-129		100	<i>Pseudomonas aeruginosa</i> (DQ236171)
		blaOXA-119		99.5	<i>Salmonella enterica</i> (AM932669)
	Fluoroquinolone resistance	qnrVC1	1	97	Uncultured bacterium (AY139598)
		Gene_699659_Scaffold_700164		100	<i>Vibrio cholerae</i> (EU436855)
	Macrolide, Lincosamide and Streptogramin B resistance	msr(D)	4	92.72	<i>Streptococcus pneumoniae</i> (AF227520)
		msr(D)		91.46	<i>Streptococcus pneumoniae</i> (AF274302)
		msr(E)		100	<i>Acinetobacter baumannii</i> (EU294228)
		msr(D)		91.48	<i>Streptococcus pneumoniae</i> (AF227520)
	Macrolide resistance	mef(B)	10	96.52	<i>Escherichia coli</i> (FJ196385)
		ere(D)		98.61	<i>Riemerella anatipestifer</i> (KP265721)
		erm(B)		100	<i>Enterococcus faecium</i> (JN899585)
		erm(X)		99.64	<i>Corynebacterium diphtheriae</i> (X51472)
		mph(F)		99.51	Uncultured bacterium (AM260957)
		ere(A)		99.48	<i>Escherichia coli</i> (DQ157752)
		erm(F)		99.63	<i>Bacteroides fragilis</i> (M17808)
		mph(E)		100	<i>Acinetobacter baumannii</i> (EU294228)
		mph(A)		99.89	<i>Escherichia coli</i> (D16251)
		erm(G)		98.83	<i>Lysinibacillus sphaericus</i> (M15332)
	Lincosamide resistance	lnu(B)	2	99.42	(AJ23824)
		lnu(F)		99.64	<i>Escherichia coli</i> (AJ561197)
	Phenicol resistance	floR	3	95.06	<i>Salmonella typhimurium</i> (AF118107)
		cmlA1		99.13	<i>Escherichia coli</i> (AB212941)
	Rifampicin resistance	catQ	2	99.7	<i>Clostridium perfringens</i> (M55620)
		ARR-6		99.77	<i>Pseudomonas aeruginosa</i> (FM897214)
	Sulphonamide resistance	ARR-2	6	100	<i>Acinetobacter baumannii</i> (HQ141279)
		sul3		99.81	<i>Escherichia coli</i> (AJ459418)
		sul3		90.97	<i>Escherichia coli</i> (AJ459418)
		sul2		100	<i>Vibrio cholerae</i> (AY034138)
		sul1		100	<i>Pseudomonas aeruginosa</i> (U12338)
		sul4		100	Uncultured bacterium (MG649393)
		sul1		99.88	<i>Klebsiella pneumoniae</i> (DQ143913)
		tet(C)		99.83	<i>Francisella tularensi</i> (AF055345)
	Tetracycline resistance	tet(W)	8	98.5	<i>Bifidobacterium longum</i> (DQ060146)
		tet(32)		100	<i>Escherichia coli</i> (EU722333)
		tet(M)		99.84	<i>Streptococcus conjugative</i> (X04388)
		tet(36)		90.83	<i>Bacteroides sp.</i> (AJ514254)
		tet(O)		99.79	<i>Campylobacter jejuni</i> (M18896)
		tet(A)		100	<i>Acinetobacter baumannii</i> (AY196695)
		tet(Q)		99.84	<i>Prevotella ruminicola</i> (L33696)
	Trimethoprim resistance	dfrA1	1	100	<i>Escherichia coli</i> (X00926)

lactam resistance), *FR* (fluoroquinolone resistance), *MR* (macrolide resistance), *ML/SB* (MLS resistance), *LR* (lincosamide resistance), *PR* (phenicol resistance), *RR* (rifampicin resistance), *SR* (sulphonamide resistance), *TER* (tetracycline resistance), and *TRR* (trimethoprim resistance) genes, four types of ARGs, namely, *AGR*, *BLR*, *MR*, and *TER*, had higher abundance values in Y1S1 and Y3S3 than in Y2S2 (Fig. 3).

3.3. Phylogenetic analyses

A total of 129 resistance genes, identified from 39 different microbial species present in the samples collected from the sediments of the Yamuna River, were considered for phylogenetic analyses. Phylogenetic

analyses revealed that, all the ARGs belonged to different microbial strains that clustered into 8 groups for their evolutionary relationships (Fig. 4). Analysis of the sub-tree of the circular cladogram underlined the fact that, the similar types of ARGs are conserved throughout evolution. It was additionally identified that the *BLR* genes in Group-1, namely, *blaOXA-10*, *blaOXA-129*, and *blaOXA-347* showed higher evolutionary conservedness with bootstrap values higher than 90%. Interestingly, the *MR* genes in Group-2, including *erm* (G) and *erm* (A) showed higher evolutionary closeness to the *BLR* genes, *blaOXA-1* and *blaOXA-4*, with the highest bootstrap percent value. The *TER* genes and *AGR* genes in Group-3 and Group-4, respectively, were clustered together with a significant bootstrap value. Among all the evolutionary

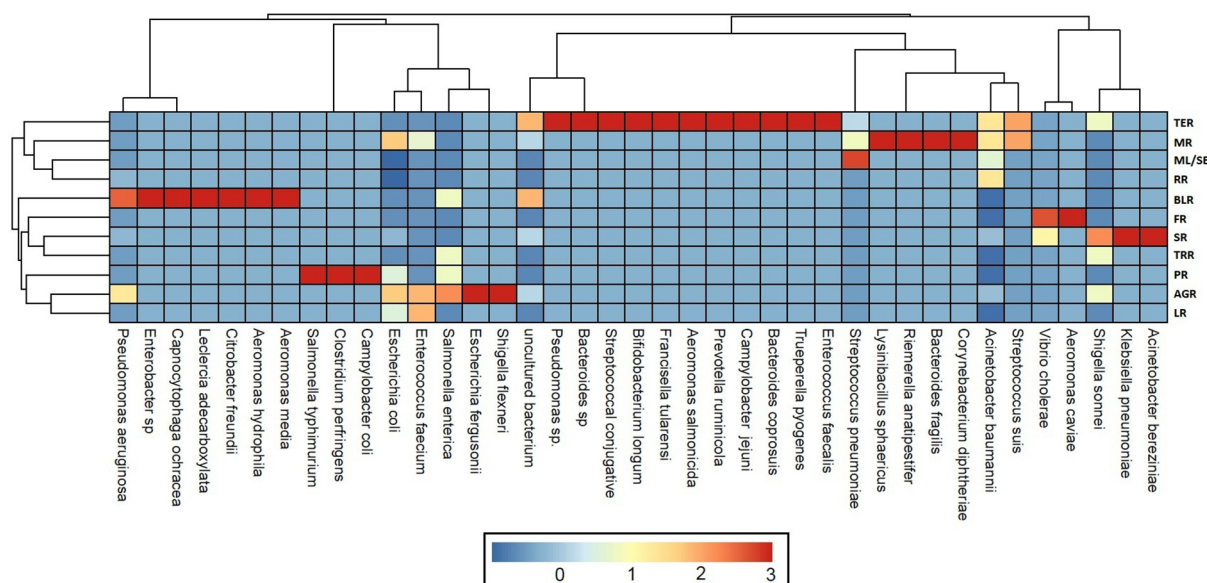


Fig. 2. Heat map representation of the number of observed resistance microbial species with respect to different types of antibiotic resistance. where AGR, BLR, FR, MR, ML/SB, LR, PR, RR, SR, TER, and TRR represent Amino glycoside resistance, Beta-lactam resistance, Fluoroquinolone resistance, Macrolide resistance, Macrolide Lincosamide and StreptograminB resistance, Lincosamide resistance, Phenicol resistance, Rifampicin resistance, Sulphonamide resistance, Tetracycline resistance and Trimethoprim resistance respectively.

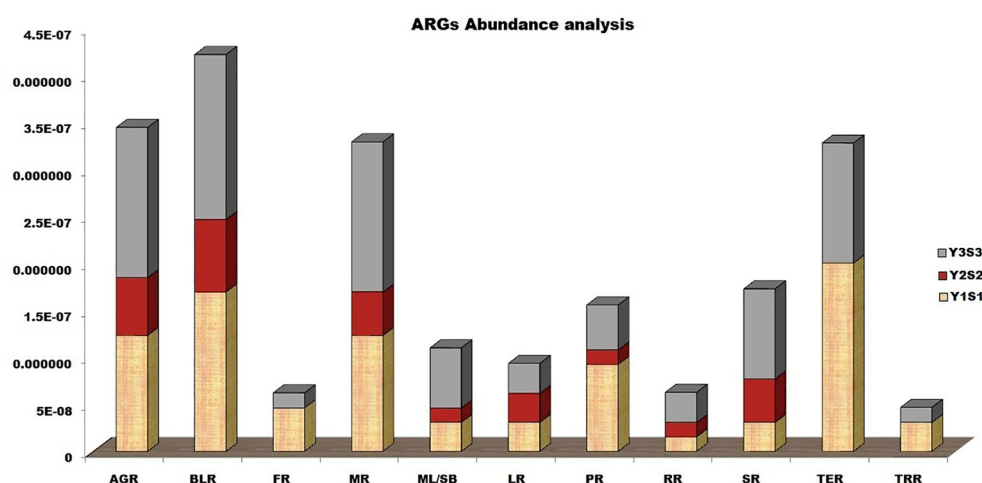


Fig. 3. Abundance of ARGs identified from the three different sites of river Yamuna. where AGR, BLR, FR, MR, ML/SB, LR, PR, RR, SR, TER, and TRR represent Amino glycoside resistance, Beta-lactam resistance, Fluoroquinolone resistance, Macrolide resistance, Macrolide Lincosamide and StreptograminB resistance, Lincosamide resistance, Phenicol resistance, Rifampicin resistance, Sulphonamide resistance, Tetracycline resistance and Trimethoprim resistance respectively.

clusters, Group-5 comprised the highest number (36) of closely related ARGs which exhibited significant phylogenetic relationships with the AGR gene *ant* (2'')-Ia, SR gene *sul2*, TER genes, *tetA* and *tetG*, and PR genes, *floR*, and *cmlA1*. Similarly, the ARGs in Group-6 and Group-7 were evolutionarily close and included *blaNPS* and *blaGES-5*, which belongs to the BLR group. Group-8 comprised 19 ARGs conferring resistance to macrolide and trimethoprim antibiotics, which clustered with a considerably high bootstrap value.

3.4. Network and functional enrichment analysis

Network analysis was performed for determining the interactions among the proteins that were predicted from the ARG-encoded ORFs, in the sediment microbiome of the Yamuna River in New Delhi. The interaction network was generated using the STRING server and is depicted in Fig. 5. We identified 160 nodes that were involved in protein-protein interactions (PPIs) with very low PPI enrichment p-value ($< 1.0 \times 10^{-16}$) and an average local clustering coefficient value of 0.732. The abbreviated names of the proteins, their description, and the number of amino acid residues are enlisted in Supplementary Table 1.

Fig. 5 shows that all the proteins in the network interacted with each other. Functional pathways enrichment analysis of the PPI network using the STRING web server revealed that 76, 36, 18, and 5 GO-terms were significantly enriched in Biological Process, Molecular Function, Cellular Component, and KEGG Pathways, respectively. Interestingly, two GO terms in Biological Process, namely, GO:0046677 (gene 15) and GO:0009636 (gene 16) had false discovery rates of 7.74e-05 and 8.44e-05, respectively, and were involved in the response to antibiotics and toxic substances.

4. Discussion

The increase in the population of antibiotic-resistant bacteria in natural ecosystems, including the aquatic environment, is a major health concern. This is augmented by the inefficient or inadequate treatment of sewage, resulting in the release of improperly treated waste materials into rivers (Meyer et al., 1994; Chandy, 2008). The MDR bacteria present in water sources are health hazards to the human community (Marathe et al., 2013). The habitat and environment play a critical role in the development of antibiotic resistance in bacteria. The

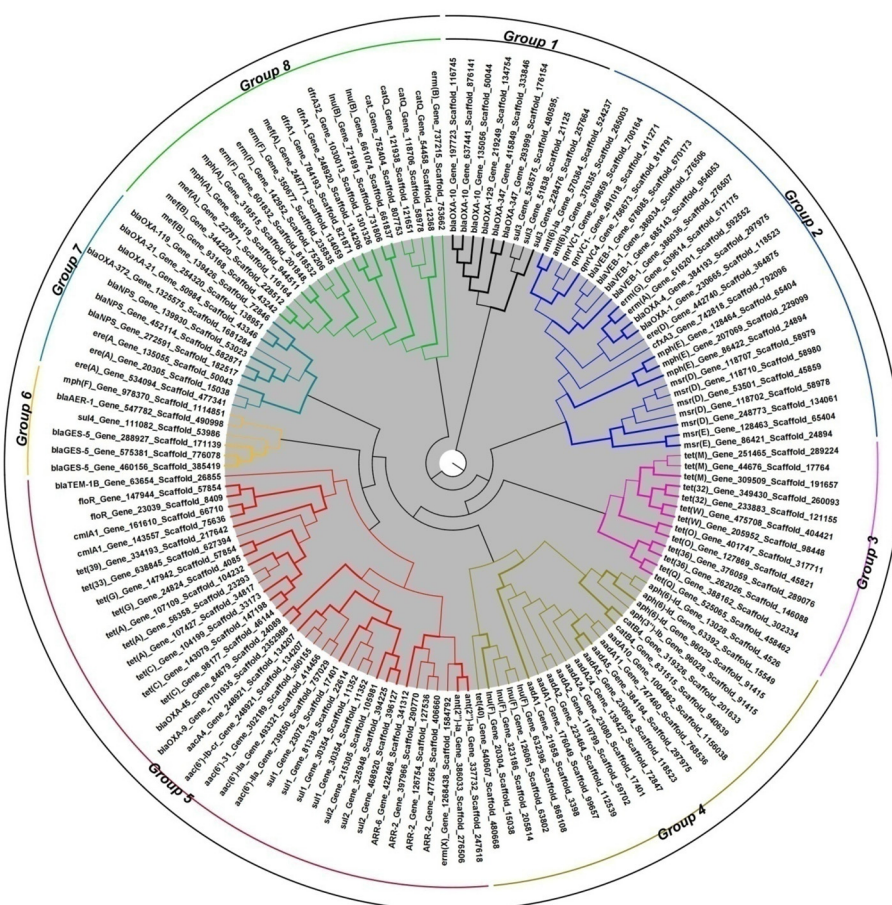


Fig. 4. Phylogenetic tree represent 129 identified antibiotic resistance genome of sediment microbes derived from the sediments of the river Yamuna. Circular cladogram visibly demarcated eight different Groups represented in different colours. The highest numbers of evolutionary closed resistance genes were found in Group 5 represented in red colour. ARGs which are closely related throughout the evolution with bootstrap value more than 90 present were highlighted and represented with bold line in the circular phylogenetic tree. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

residual antibiotics in wastewater are pollutants that exert selection pressure and enhance the development of antibiotic resistance in the bacterial species present in aquatic environments (Li et al., 2010). Sewage from hospitals and human habitats, agricultural effluents, and wastewater from aquaculture farms are also important sources of residual antibiotics, which leads to the emergence of drug resistant bacteria in polluted water sources (Li et al., 2010; Prado et al., 2017). Additionally, the direct discharge of domestic sewage into open water bodies without appropriate pre-treatment contributes to the development of MDR strains, enabling the bacteria to adapt and survive.

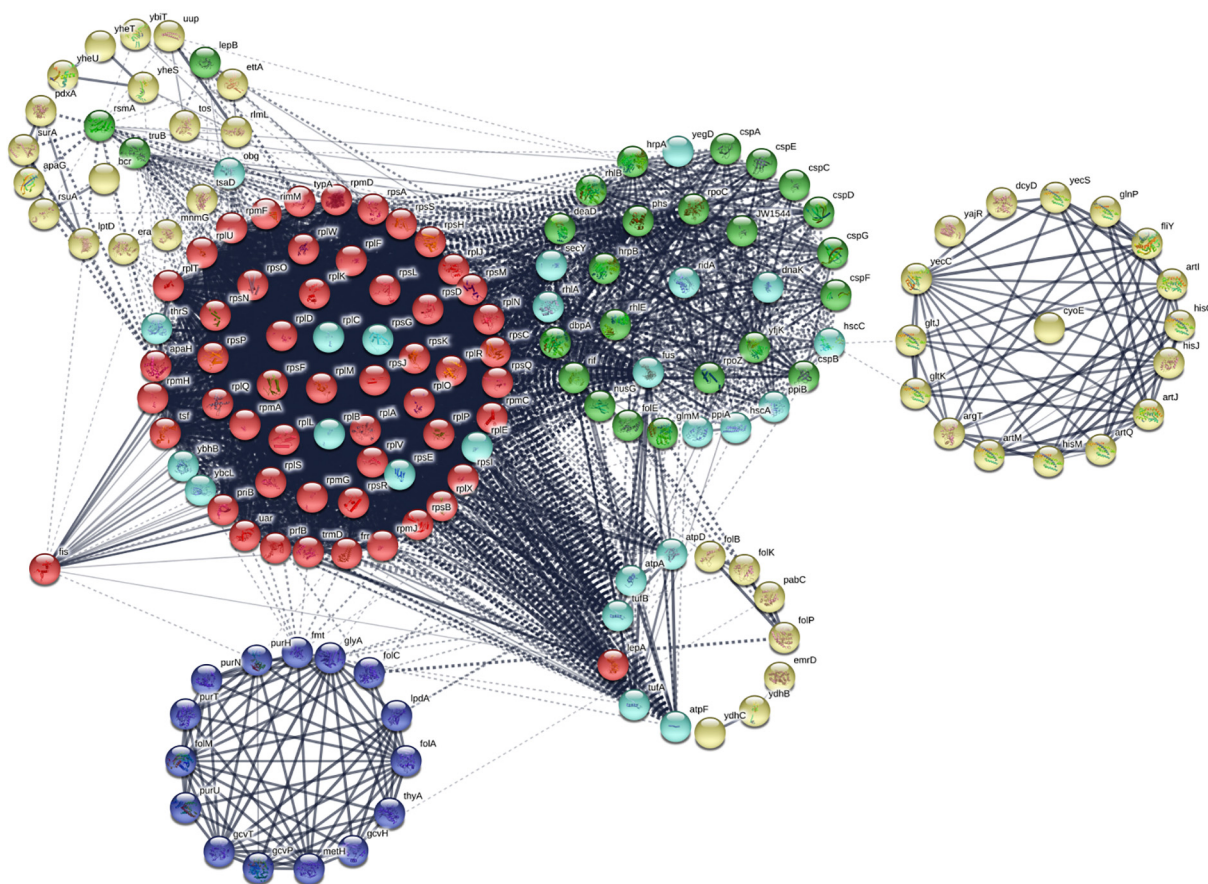
As the prevention of MDR bacterial infections relies on the precise detection of MDR strains, the present study aimed to identify ARGs from sediment samples of the Yamuna River using a metagenomics approach (Andersen et al., 2016). A sediment metagenomic study was designed, for identifying and studying the ARGs in the microbial community present in the Yamuna River. In this study, a total of 139 ARGs, conferring resistance to 11 antibiotics, were identified in 39 different microbial species present in the sediment samples collected from three different sites along the Yamuna River using a metagenomic approach.

Metagenomic analyses revealed that, the resistome identified from the sediments of the Yamuna River showed resistance against 11 different types of antibiotics, namely, aminoglycoside, beta-lactam, fluoroquinolone, macrolide, macrolide lincosamide, and streptogramin B, lincosamide, phenicol, rifampicin, sulphonamide, tetracycline, and trimethoprim antibiotics. A report on “Antimicrobial Resistance in India” published by in 2017 by DBT, New Delhi, India, highlighted the increasing load of antibiotic-resistant bacteria and ARGs in the surface water bodies and major drinking water sources of India (DBT, 2017). Numerous culture-dependent studies have also been undertaken for assessing the presence of antibiotic-resistant bacteria in the surface and

groundwater sources across India (Kumar et al., 2013; Poonia et al., 2014). The presence of MDR bacteria has been reported in the major Indian rivers, including the Ganges, the Yamuna, and Cauvery (Ram et al., 2007; Skariyachan et al., 2015; Azam et al., 2016; Lamba et al., 2017; Mittal et al., 2019). A separate study by Mittal and coworkers in 2019 also reported the presence broad-spectrum antibiotics, including carbapenemases and metallo- β -lactamases, in the Yamuna River. The occurrence of ARGs and antibiotic resistant bacteria in different riverine ecosystems of India has been reported by several workers. The tetracycline, kanamycin, and sulfonamide resistant genes are more prevalent in the Sabarmati River (Ram and Kumar, 2020), while beta-lactam resistant genes was reported from the Cauvery and Kshipra rivers (Devarajan et al., 2016). However, genes resistant against levofloxacin, ciprofloxacin, norfloxacin, kanamycin, tetracycline, and sulfamethoxazole are predominantly present in the Brahmaputra River (Chitnis et al., 2004).

ARGs against Tetracycline, Kanamycin, sulphonamide, Levofloxacin, Ciprofloxacin, and Norfloxacin are common in the Indian River system. However, the percentage, abundance, diversity, and geographical distribution of ARGs is highly dynamic and changes with seasonal and geographical variations. A higher concentration of ARGs was found in more urbanized sites rather than the upstream condition correlating the anthropogenic activities and higher resistance percentage (Honda et al., 2016; Ram and Kumar, 2020; Kumar et al., 2020).

The presence of ARGs in the environment is not restricted to a particular geographical location. The occurrence of ARGs in freshwater environments, including streams, lakes, rivers, are focused mostly around the world (Nnadozie and Odume, 2019) and high relative abundances of ARGs in seawater plastic samples have been studied in details (Yang et al., 2019). A total of 23 ARGs were investigated which includes sulfonamide, β -lactam, chloramphenicol, erythromycin, and



tetracycline resistance genes. Sulfonamide resistance genes were the most frequently detected ARGs in the Beiji River in China (Jiang et al., 2018). Similarly, nine different ARGs, including tetracycline, fluoroquinolone, and sulfonamide resistance genes were detected in the Weihe River. Tetracycline and fluoroquinolone were frequently detected in all the samples collected from the Weihe River (Li et al., 2010). Kumar et al. (2020) also detected that the percentage of resistant genes against tetracycline, sulfonamides, β -lactams, and fluoroquinolones are very high in the Kelani River of Sri Lanka. Similar studies performed with samples from the Chaophraya River in Thailand revealed the presence of high percentages of fluoroquinolones (Honda et al., 2016). A river in the Mediterranean region showed the presence of six clinically relevant ARGs, *blaTEM*, *blaCTX-M*, *qnrA*, *qnrS*, *mecA*, and *sul1*. Of these, *blaTEM* and *sul1* were the most prevalent and abundant ARGs (Calero-Cáceres et al., 2017). Furthermore, ARGs conferring resistance against sulfonamide, tetracycline, and macrolide antibiotics have also been reported from the Poudre River in Colorado (Garner et al., 2016). In 2018, Sabri and co-workers (Sabri et al., 2020) identified several ARGs, including *ermB*, *sul1*, *sul2*, and *tetW*, from the sediment samples collected from the Grote Beerze River in the Netherlands. In 2018, Gonzalez-Plaza and co-workers reported eight potentially novel ARGs, including those conferring resistance to dihydrofolate reductase and beta-lactamase antibiotics, belonging to classes A, B, and D, in the sediment samples obtained from the Sava River in Croatia, Europe (González-Plaza et al., 2018).

confering resistance to carbapenemases, such as *blaNDM*, *blaVIM*, *blaKPC*, *blaOXA-48* and *blaIMP*. In 2018, Marathe and co workers (Marathe et al., 2018) identified seven novel resistance gene candidates, including six *BLR* genes and one amikacin resistance gene from the sediments of the Isakavagu River in India, that is polluted with fluoroquinolones and other drugs that are discharged into it. The present study reports the presence of MDR bacteria, including *P. aeruginosa*, *S. sonnei*, *E. coli*, *E. faecium*, *V. cholerae*, *S. pneumoniae*, *A. baumannii*, *S. suis*, and *S. enteric*, in addition to a few uncultured bacteria in the sediment samples collected from three sites along the Yamuna River. Relative abundance analysis revealed that the *AGR*, *BLR*, *MR*, and *TER* genes showed a higher abundance in the Y1S1 and Y3S3 samples, than in the Y2S2 sample, similar to the observation reported by Mittal and coworkers in 2019.

At present, the evolution of ARGs in environmental microbes and their spread is one of the most threatening global health care problems (Hawkey, 2008). In this study, the evolution of ARGs in the identified resistome was investigated by phylogenetic analyses based on multiple sequence alignment. Phylogenetic analysis clearly revealed the diversity of the resistome, and the ARGs identified herein were clustered into eight diverse groups. Closer inspection of the first phylogenetic group (Group-1) revealed that *blaOXA-10*, *blaOXA-129*, and *blaOXA-347* had a close evolutionary relationship, with bootstrap values higher than 90%. Interestingly, the *MR* genes (*ermG* and *ermA*) in the second phylogenetic group (Group-2) showed the highest bootstrap value with the *BLR* genes *blaOXA-1* and *blaOXA-4*, signifying a strong phylogenetic relationship between these two classes of ARGs. In 2006, Bergman and co-workers reported (Bergman et al., 2006) that among the 31,000 strains of *Pneumococcus* identified from Finland, the *MR* genes and

ARGs conferring resistance to previous macrolides or azithromycin has a strong association with ARGs conferring resistance to previous beta-lactam antibiotics, cephalosporin, or penicillin. Among all the phylogenetic clusters, Group-5 consisted of the highest number (36) of closely related ARGs which had a significantly strong phylogenetic relationship with the ARGs *ant(2'')-Ia*, *SR* gene *sul2*, *TER* genes *tetA* and *tetG*, and the *PR* genes *floR* and *cmlA1*. Samson and coworkers identified genes conferring resistance to antibiotics such as beta lactams and fluoroquinolones in the sediment samples of the Yamuna River (2019).

In this study, the study aimed to understand the functional assignment by analysing the interactions among the proteins encoded by the predicted ARG-encoded ORFs in the sediment microbiome of the Yamuna River. A total of 160 nodes were identified by PPI analysis, which was involved in network formation. All the proteins in the network were found to be integrated to one other. Functional pathway enrichment analysis of the network was performed using the STRING web server, which revealed significantly enriched GO-terms such as Biological process, Molecular Function, Cellular Component, and KEGG Pathways, respectively (Supplementary Table S1). Interestingly, two GO terms in Biological Process, namely GO:0046677 and GO:0009636, were found to be involved in the response to antibiotics and toxic substances. The interaction network generated in the present study would help understand the various mechanisms underlying the antibiotic resistance and the associated gene or protein interactions of the identified ARGs. These findings provide deeper insight into the mechanisms underlying the antibiotic resistance identified in the sediment resistome of the Yamuna River and elucidate the interactions between the ARGs and their functional partners.

5. Conclusions

The random use of antibiotics in agriculture and human health leads to the accumulation of antibiotics in the environment, which acts as a driving force for the development of antibiotic-resistant pathogenic microbial strains. In this study, the profiles of the ARGs that were identified from the riverine ecosystem of the Yamuna by analysing samples collected from three different sites near New Delhi, India, were comprehensively characterized and compared using a high throughput screening-based metagenomic approach. A total of 11 different ARGs were identified in the sediment metagenome of the Yamuna River, of which the *AGR*, *BLR*, *MR*, and *TER* genes were found to be highly abundant. The abundance of different antibiotic resistant genes identified in the present study was high in sediment sample collected from Wazaribad and Faizupur Khaddar in comparison to sediment sample collected from Okhla barrage though the physicochemical property of water sample at Wazaribad is little better than water from Okhla barrage. In contradiction to our study Lamba et al. (2017) found less abundance of ARGs (TC, FC, ESBL) in sediment samples collected from Wazaribad. The results of this study added to the information present in the existing literature on the scenario of ARGs in the riverine ecosystems of India. Phylogenetic analysis of the identified ARGs, based on multiple sequence alignment, demonstrated the diversity of the resistome of the Yamuna River sediments. The ARGs clustered into eight diverse groups. Analysis of the ARG interaction network provided novel information pertaining to the various mechanisms of drug resistance, and elucidated the interactions between the ARGs and the associated genes or proteins. The findings of this study will aid researchers in understanding the mechanisms underlying the antimicrobial resistance (AMR) of the O157:H7 strain of *E. coli* and elucidate the interactions between the ARGs and their functional partners. Furthermore, the information provided herein will aid the development of next generation antibiotics and encourage proper actions for the management of sewage, agricultural waste, and hospital effluents that are discharged into the riverine ecosystems of India.

CRedit authorship contribution statement

Basanta Kumar Das: Conceptualization, Writing - review & editing. **Bijay Kumar Behera:** Investigation, Methodology, Funding acquisition, Writing - review & editing. **Hirak Jyoti Chakraborty:** Data curation, Formal analysis. **Prasenjit Paria:** Writing - original draft. **Aditi Gangopadhyay:** Formal analysis. **Ajaya Kumar Rout:** Formal analysis. **Kausalya Kumari Nayak:** Writing - original draft. **Pranaya Kumar Parida:** Project administration, Investigation, Methodology. **Anil Rai:** Funding acquisition, Writing - review & editing.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Acknowledgements

This work has been financially supported by the ICAR CABIN Project, New Delhi, India. Authors are thankful to Dr. J. K. Jena, Deputy Director General (Fisheries Science), Indian Council of Agricultural Research, New Delhi for his guidance and support. The authors express their gratitude to Mr. Asim Kumar Jana, Senior Technical Assistant for his aid during sampling and laboratory assistance.

Appendix A. Supplementary data

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.gene.2020.144951>.

References

- Altschul, S.F., Gish, W., Miller, W., Myers, E.W., Lipman, D.J., 1990. Basic local alignment search tool. *J. Mol. Biol.* 215, 403–410.
- Andersen, H., Connolly, N., Bangar, H., Staat, M., Mortensen, J., Deburger, B., Haslam, D.B., 2016. Use of shotgun metagenome sequencing to detect fecal colonization with multidrug-resistant bacteria in children. *J. Clin. Microbiol.* 54, 1804–1813.
- Azam, M., Jan, A.T., Haq, Q.M., 2016. blaCTX-M-152, a novel variant of CTX-M-group-25, identified in a study performed on the prevalence of multidrug resistance among natural inhabitants of river Yamuna, India. *Front. Microbiol.* 7, 176.
- Bengtsson-Palme, J., Boulund, F., Fick, J., Kristiansson, E., Larsson, D.G., 2014. Shotgun metagenomics reveals a wide array of antibiotic resistance genes and mobile elements in a polluted lake in India. *Front. Microbiol.* 5, 648.
- Bergman, M., Huikko, S., Huovinen, P., Paakkari, P., Seppälä, H., Renkonen, R., Muotiala, A., Vaara, M., Carlson, P., Somer, H., Virolainen-Julkunen, A., 2006. Macrolide and azithromycin use are linked to increased macrolide resistance in *Streptococcus pneumoniae*. *Antimicrob. Agents. Chemother.* 50, 3646–3650.
- Calero-Cáceres, W., Méndez, J., Martín-Díaz, J., Muniesa, M., 2017. The occurrence of antibiotic resistance genes in a Mediterranean river and their persistence in the riverbed sediment. *Environ. Pollut.* 223, 384–394.
- Chandy, S.J., 2008. Consequences of irrational use of antibiotics. *Indian J. Med. Ethics.* 5, 174–175.
- Chen, B., Lin, L., Fang, L., Yang, Y., Chen, E., Yuan, K., Zou, S., Wang, X., Luan, T., 2018. Complex pollution of antibiotic resistance genes due to beta-lactam and aminoglycoside use in aquaculture farming. *Water Res.* 134, 200–208.
- Chitnis, V., Chitnis, S., Vaidya, K., Ravikant, S., Patil, S., Chitnis, D.S., 2004. Bacterial population changes in hospital effluent treatment plant in central India. *Water Res.* 38, 441–447.
- D'Costa, V.M., King, C.E., Kalan, L., Morar, M., Sung, W.W., Schwarz, C., Froese, D., Zazula, G., Calmels, F., Debruyne, R., Golding, G.B., 2011. Antibiotic resistance is ancient. *Nature* 477, 457–461.
- DBT, 2017 Scoping Report on Antimicrobial Resistance in India; Department of Biotechnology: New Delhi, India.
- Devarajan, N., Laffite, A., Mulaji, C.K., Otamanga, J.P., Mpiana, P.T., Mubedi, J.I., Prabakar, K., Ibelings, B.W., Poté, J., 2016. Occurrence of antibiotic resistance genes and bacterial markers in a tropical river receiving hospital and urban wastewaters. *PLoS One* 11 (2), e0149211.
- Franceschini, A., Szklarczyk, D., Frankild, S., Kuhn, M., Simonovic, M., Roth, A., Lin, J., Minguez, P., Bork, P., Von Mering, C., Jensen, L.J., 2012. STRING v9.1: protein-protein interaction networks, with increased coverage and integration. *Nucl. Acids. Res.* 41, 808–815.
- Garner, E., Wallace, J.S., Argoty, G.A., Wilkinson, C., Fahrenfeld, N., Heath, L.S., Zhang, L., Arabi, M., Aga, D.S., Pruden, A., 2016. Metagenomic profiling of historic Colorado Front Range flood impact on distribution of riverine antibiotic resistance genes. *Sci.*

- Rep. 6, 38432.
- González-Plaza, J.J., Šimatović, A., Milaković, M., Bielen, A., Wichmann, F., Udiković-Kolić, N., 2018. Functional repertoire of antibiotic resistance genes in antibiotic manufacturing effluents and receiving freshwater sediments. *Front. Microbiol.* 8, 2675.
- Guo, J., Li, J., Chen, H., Bond, P.L., Yuan, Z., 2017. Metagenomic analysis reveals wastewater treatment plants as hotspots of antibiotic resistance genes and mobile genetic elements. *Water. Res.* 123, 468–478.
- Hawkey, P.M., 2008. The growing burden of antimicrobial resistance. *J. Antimicrob. Chemother.* 62, i1–i9.
- Honda, R., Watanabe, T., Sawaitayotin, V., Masago, Y., Chulasak, R., Tanong, K., Chaminda, G.T., Wongsila, K., Sienglum, C., Sunthonwathanaphong, V., Poonnotok, A., 2016. Impacts of urbanization on the prevalence of antibiotic-resistant *Escherichia coli* in the Chaophraya River and its tributaries. *Water Sci. Technol.* 73, 362–374.
- Jiang, H., Zhou, R., Yang, Y., Chen, B., Cheng, Z., Zhang, M., Li, J., Zhang, G., Zou, S., 2018. Characterizing the antibiotic resistance genes in a river catchment: Influence of anthropogenic activities. *J. Environ. Sci.* 69, 125–132.
- Kumar, S., Tripathi, V.R., Garg, S.K., 2013. Antibiotic resistance and genetic diversity in water-borne Enterobacteriaceae isolates from recreational and drinking water sources. *Int. J. Environ. Sci. Technol.* 10, 789–798.
- Kumar, M., Chaminda, G.T., Honda, R., 2020a. Seasonality impels the antibiotic resistance in Kelani River of the emerging economy of Sri Lanka. *npj Clean Water* 3 (1), 1–8.
- Lamba, M., Sreekrishnan, T.R. and Ahammad, S.Z., 2017. Sewage mediated transfer of antibiotic resistance to River Yamuna in Delhi, India. *J. Environ. Chem. Eng.* 102088.
- Li, D., Yu, T., Zhang, Y., Yang, M., Li, Z., Liu, M., Qi, R., 2010. Antibiotic resistance characteristics of environmental bacteria from an oxytetracycline production wastewater treatment plant and the receiving river. *Appl. Environ. Microbiol.* 76, 3444–3451.
- Kumar, M., Chaminda, G.G.T., Honda, R., 2020b. Seasonality impels the antibiotic resistance in Kelani River of the emerging economy of Sri Lanka. *npj Clean Water* 3 (1), 1–8.
- Marathe, N.P., Regina, V.R., Walujkar, S.A., Charan, S.S., Moore, E.R., Larsson, D.J., Shouche, Y.S., 2013. A treatment plant receiving waste water from multiple bulk drug manufacturers is a reservoir for highly multi-drug resistant integron-bearing bacteria. *PLoS One* 8, e77310.
- Marathe, N.P., Janzon, A., Kotsakis, S.D., Flach, C.F., Razavi, M., Berglund, F., Kristiansson, E., Larsson, D.J., 2018. Functional metagenomics reveals a novel carbapenem-hydrolyzing mobile beta-lactamase from Indian river sediments contaminated with antibiotic production waste. *Environ. Int.* 112, 279–286.
- Marathe, N.P., Pal, C., Gaikwad, S.S., Jonsson, V., Kristiansson, E., Larsson, D.J., 2017. Untreated urban waste contaminates Indian river sediments with resistance genes to last resort antibiotics. *Water. Res.* 124, 388–397.
- Martínez, J.L., 2008. Antibiotics and antibiotic resistance genes in natural environments. *Science* 321, 365–367.
- Metsalu, T., Vilo, J., 2015. ClustVis: a web tool for visualizing clustering of multivariate data using Principal Component Analysis and heatmap. *Nucl. Acids. Res.* 43, W566–W570.
- Meyer, C.N., Skinhøj, P., Prag, J., 1994. Bacteremia in HIV-positive and AIDS patients: incidence, species distribution, risk-factors, outcome, and influence of long-term prophylactic antibiotic treatment. *Scand. J. Infect. Dis.* 26, 635–642.
- Misra, A.K., 2010. A river about to die: Yamuna. *J. Water. Resource. Prot.* 2, 489.
- Mittal, P., PK, V.P., Dhakan, D.B., Kumar, S. and Sharma, V.K., 2019. Metagenome of a polluted river reveals a reservoir of metabolic and antibiotic resistance genes. *Environmental Microbiome.* 14, 5.
- Mullany, P., 2014. Functional metagenomics for the investigation of antibiotic resistance. *Virulence* 5, 443–447.
- Nnadozie, C.F., Odume, O.N., 2019. Freshwater environments as reservoirs of antibiotic resistant bacteria and their role in the dissemination of antibiotic resistance genes. *Environ. Pollut.*, 113067.
- O'Neill, J., 2018. Tackling drug-resistant infections globally: Final report and recommendations. 2016. HM Government and Wellcome Trust: UK.
- O'Neill, J., 2016. Tackling drug-resistant infections globally: Final report and recommendations. Review on Antimicrobial Resistance.
- Poonia, S., Singh, T.S., Tsering, D.C., 2014. Antibiotic susceptibility profile of bacteria isolated from natural sources of water from rural areas of East Sikkim. *Indian J. Community. Med.* 39, 156.
- Prado, T., Bruni, A.D.C., Barbosa, M.R.F., Bonanno, V.M.S., Garcia, S.C., Sato, M.I.Z., 2017. Distribution of human fecal marker gb-124 bacteriophages in urban sewage and reclaimed water of São Paulo city, Brazil. *J. Water Health* 16, 289–299.
- Ram, S., Vajpayee, P., Shanker, R., 2007. Prevalence of multi-antimicrobial-agent resistant, shiga toxin and enterotoxin producing *Escherichia coli* in surface waters of river Ganga. *Environ. Sci. Technol.* 41, 7383–7388.
- Ram, B., Kumar, M., 2020. Correlation appraisal of antibiotic resistance with fecal, metal and microplastic contamination in a tropical Indian river, lakes and sewage. *NPJ Clean Water* 3 (1), 1–12.
- Sabri, N.A., Schmitt, H., Van der Zaan, B., Gerritsen, H.W., Zuidema, T., Rijnaarts, H.H.M., Langenhoff, A.A., 2020. Prevalence of antibiotics and antibiotic resistance genes in a wastewater effluent-receiving river in the Netherlands. *J. Environ. Chem. Eng.* 8 (1), 102245.
- Sahu, V., Sohoni, P., 2014. Water quality analysis of river Yamuna-the Delhi stretch. *Int. J. Environ. Sci.* 4, 1177–1189.
- Samson, R., Shah, M., Yadav, R., Sarode, P., Rajput, V., Dastager, S.G., Dharne, M.S., Khairnar, K., 2019. Metagenomic insights to understand transient influence of Yamuna River on taxonomic and functional aspects of bacterial and archaeal communities of River Ganges. *Sci. Total Environ.* 674, 288–299.
- Skariyachan, S., Mahajanakatti, A.B., Grandhi, N.J., Prasanna, A., Sen, B., Sharma, N., Vasist, K.S., Narayanappa, R., 2015. Environmental monitoring of bacterial contamination and antibiotic resistance patterns of the fecal coliforms isolated from Cauvery River, a major drinking water source in Karnataka, India. *Environ. Monit. Assess.* 187, 279.
- Szklarczyk, D., Franceschini, A., Kuhn, M., Simonovic, M., Roth, A., Minguez, P., Doerks, T., Stark, M., Muller, J., Bork, P., Jensen, L.J., 2010. The STRING database in 2011: functional interaction networks of proteins, globally integrated and scored. *Nucl. Acids. Res.* D561–D568.
- Szklarczyk, D., Morris, J.H., Cook, H., Kuhn, M., Wyder, S., Simonovic, M., Santos, A., Doncheva, N.T., Roth, A., Bork, P., Jensen, L.J., 2016. The STRING database in 2017: quality-controlled protein–protein association networks, made broadly accessible. *Nucl. Acids. Res.* 937.
- Tello, A., Austin, B., Telfer, T.C., 2012. Selective pressure of antibiotic pollution on bacteria of importance to public health. *Environ. Health. Perspect.* 120, 1100–1106.
- World Health Organization, 2017. Global Priority List of Antibiotic-Resistance Bacteria to Guide Research, Discovery, and Development of New Antibiotics. World Health Organization, Geneva.
- Xu, J., Xu, Y., Wang, H., Guo, C., Qiu, H., He, Y., Zhang, Y., Li, X., Meng, W., 2015. Occurrence of antibiotics and antibiotic resistance genes in a sewage treatment plant and its effluent-receiving river. *Chemosphere* 119, 1379–1385.
- Yang, Y., Liu, G., Song, W., Ye, C., Lin, H., Li, Z., Liu, W., 2019. Plastics in the marine environment are reservoirs for antibiotic and metal resistance genes. *Environ. Int.* 123, 79–86.
- Zankari, E., Hasman, H., Cosentino, S., Vestergaard, M., Rasmussen, S., Lund, O., Aarestrup, F.M., Larsen, M.V., 2012. Identification of acquired antimicrobial resistance genes. *J. Antimicrob. Chemother.* 67, 2640–2644.
- Zeng, X., Lin, J., 2017. Factors influencing horizontal gene transfer in the intestine. *Anim. Health. Res. Rev.* 18, 153–159.